

Factory Reallocation Optimization using Machine Learning for Supply Chain Logistics

Predictive Lead Time Analysis, Factory Recommendation
System, and Interactive Decision Support Dashboard

Abstract

Efficient factory allocation is a critical aspect of supply chain management, directly affecting delivery performance and operational costs. This study presents a **machine learning-based Factory Reallocation and Shipping Optimization System** that predicts shipment lead time and recommends the most suitable manufacturing factory for each order.

The proposed system uses **Gradient Boosting Regression** to model the relationship between product attributes, factory location, shipping mode, sales, cost, profit, and shipping distance. A What-If Simulation module enables users to evaluate alternative factory assignments before implementation by recalculating shipping distances and predicting the resulting lead time.

The system supports data-driven decision-making, helping organizations reduce delivery delays, improve logistics efficiency, and optimize factory utilization. Experimental results demonstrate that the proposed approach provides accurate lead time predictions and practical recommendations for supply chain optimization.

1. Introduction

Supply chain analytics enables organizations to make data-driven decisions that improve operational efficiency, reduce transportation costs, and enhance customer satisfaction. One of the key challenges in logistics is **factory allocation**, where products must be assigned to the most suitable manufacturing facility while considering factors such as shipping distance, transportation mode, and regional demand. Efficient factory allocation directly influences delivery performance and overall supply chain effectiveness.

Another critical aspect of supply chain management is **lead time prediction**, which estimates the time required to deliver products from factories to customers. Accurate lead time prediction helps businesses optimize inventory, improve production planning, reduce delays, and provide reliable delivery commitments. These improvements lead to better resource utilization and increased customer satisfaction.

Recent advancements in **machine learning** have significantly improved logistics decision-making by identifying complex relationships within large datasets. In this project, a **Gradient Boosting Regressor** is used to predict shipment lead time based on operational and geographical features. The model supports a What-If simulation and factory reallocation process, enabling organizations to evaluate alternative factory assignments and make informed, data-driven logistics decisions.

2. Problem Statement

Many manufacturing companies allocate products to factories using predefined rules instead of data-driven decision-making. This often leads to inefficient factory assignments, longer shipping distances, and increased delivery lead times.

Accurate **lead time prediction** is essential for improving logistics planning, reducing transportation costs, and ensuring timely product delivery. However, traditional allocation methods do not effectively utilize historical operational data to support these decisions.

This research proposes a **Machine Learning-based Factory Reallocation System** using the **Gradient Boosting Regressor** to predict shipment lead time and evaluate alternative factory assignments. The objective is to identify factory reallocations that minimize lead time and improve overall supply chain efficiency.

3. Objectives

- Predict shipping lead time using machine learning.
- Compare multiple regression algorithms.
- Recommend better factories.
- Optimize transportation distance.
- Build a What-If Simulation.
- Develop an interactive dashboard.

4. Dataset Description

The dataset used in this research is the **Nassau Candy Distributor Dataset**, which contains historical shipment and order information for multiple candy products distributed across different regions of the United States. The dataset was preprocessed to include engineered features such as shipping distance and scaled numerical variables for machine learning.

Dataset Information	Details
Source	Nassau Candy Distributor Dataset
Number of Records	1,527 shipment records (<i>replace with your exact count if different</i>)
Number of Features	30 features
Product Categories	16 candy products
Factories	5 manufacturing factories
Regions	4 sales regions (Atlantic, Gulf, Interior, Pacific)

Dataset Attributes

Attribute	Description
Product Name	Candy product being shipped
Factory	Manufacturing plant assigned to produce the product
Ship Mode	Shipping method used for delivery
Division	Business division handling the order
Region	Customer destination region
Sales	Revenue generated from the order
Units	Number of units sold
Cost	Manufacturing cost of the product
Gross Profit	Profit earned from the order
Lead Time	Target variable representing delivery time (days)
Customer Coordinates	Customer latitude and longitude
Factory Coordinates	Factory latitude and longitude
Shipping Distance (km)	Distance between factory and customer calculated using Geodesic Distance
Scaled Features	Normalized values of Sales, Units, Cost, Gross Profit, and Shipping Distance used for model training

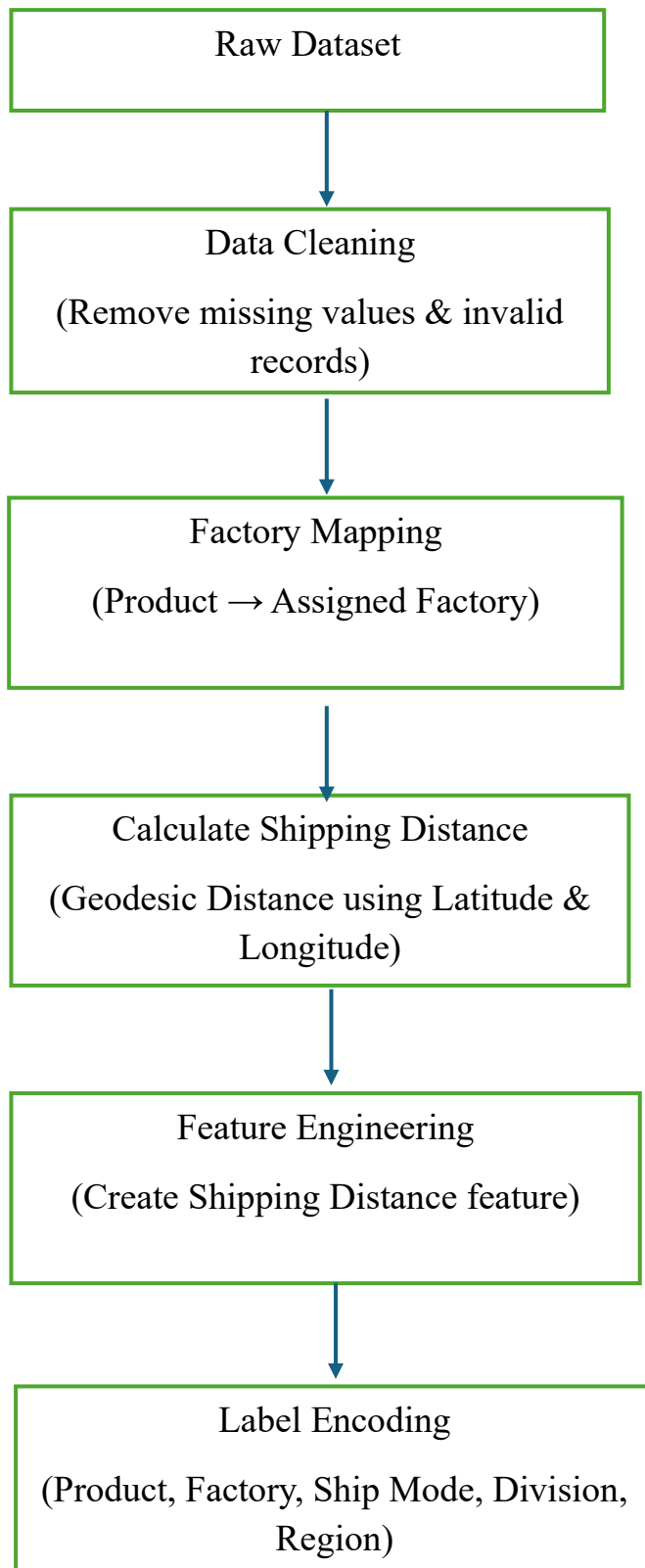
5. Data Preprocessing

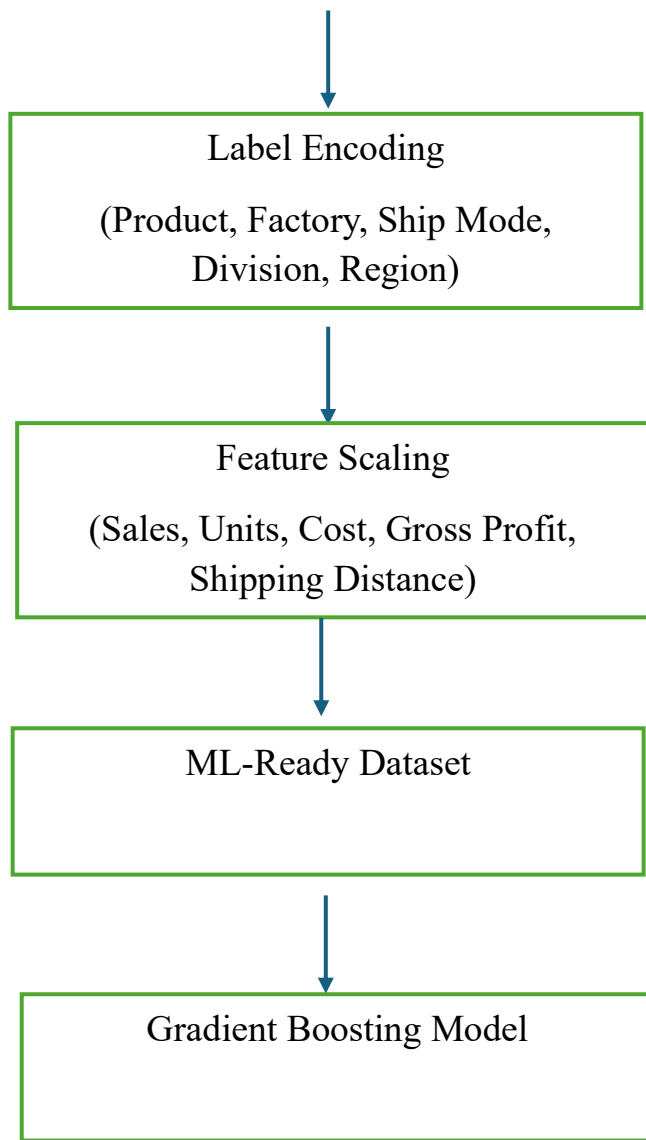
The raw shipment dataset was preprocessed to improve data quality and prepare it for machine learning. Missing values were identified and handled appropriately, while records with invalid dates or incomplete factory information were removed. This ensured that the dataset was clean and suitable for analysis.

Categorical features such as **Product Name**, **Factory**, **Ship Mode**, **Division**, and **Region** were converted into numerical values using **Label Encoding**. Numerical features including **Sales**, **Units**, **Cost**, **Gross Profit**, and **Shipping Distance** were normalized using **StandardScaler** to ensure that all features contributed equally during model training.

Additional feature engineering was performed to enhance prediction accuracy. Products were mapped to their respective factories, shipping distance was calculated using the **Geodesic Distance** formula based on factory and customer coordinates, and the calculated distance was scaled before being included in the final machine learning dataset.

Data Preprocessing Pipeline





6. Machine Learning Methodology

Three machine learning models—**Linear Regression**, **Random Forest**, and **Gradient Boosting**—were developed to predict shipment lead time. The models were trained using the preprocessed dataset and their performance was compared.

Model performance was evaluated using **RMSE**, **MAE**, and **R² Score**. Among the evaluated models, **Gradient Boosting** was selected as the final model due to its superior predictive performance and was used for lead time prediction and factory reallocation analysis.

Models Evaluated

- Linear Regression
- Random Forest
- **Gradient Boosting (Final Model)**

Evaluation Metrics

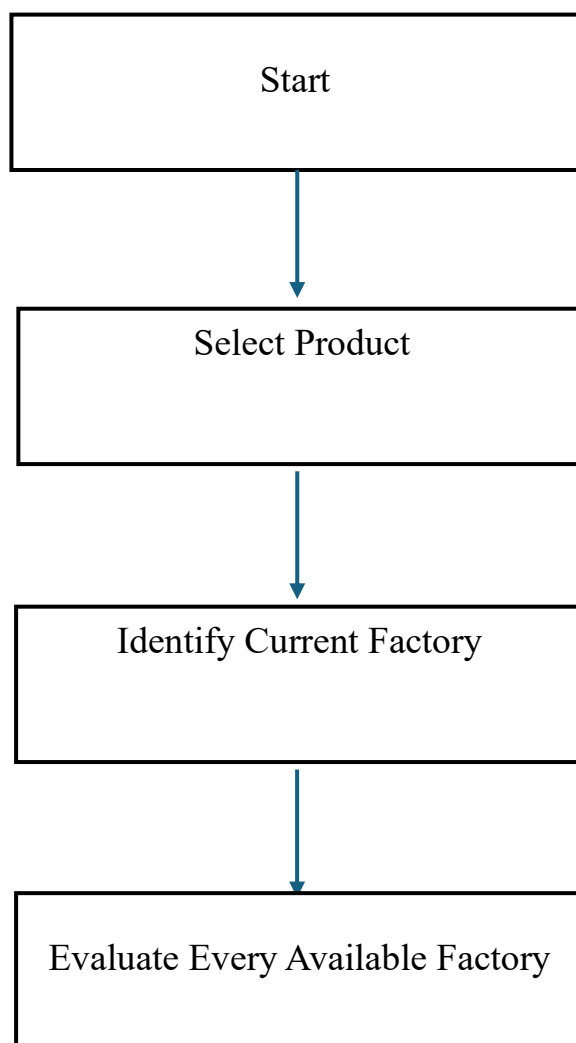
- RMSE
- MAE
- R² Score

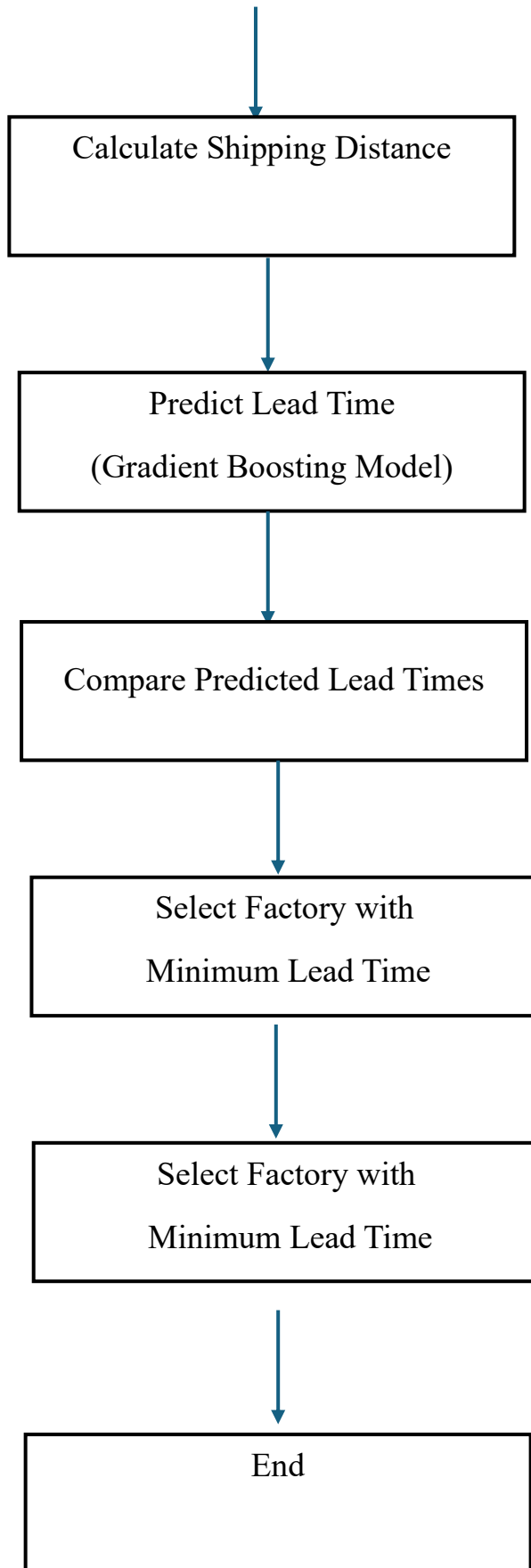
Model	RMSE	MAE	R2
Gradient Boosting	0.150373	0.128725	0.994589
Random Forest	0.152951	0.128469	0.994402
Linear Regression	0.468249	0.337618	0.947529

7. Factory Recommendation System

A factory recommendation system was developed to identify the most efficient manufacturing location for each product. Instead of relying on the current factory assignment, the system evaluates all available factories and predicts the expected lead time for each alternative using the trained Gradient Boosting model. The factory with the minimum predicted lead time is selected as the recommended production location.

This approach enables data-driven factory reallocation, helping organizations reduce delivery lead time and improve overall logistics efficiency.

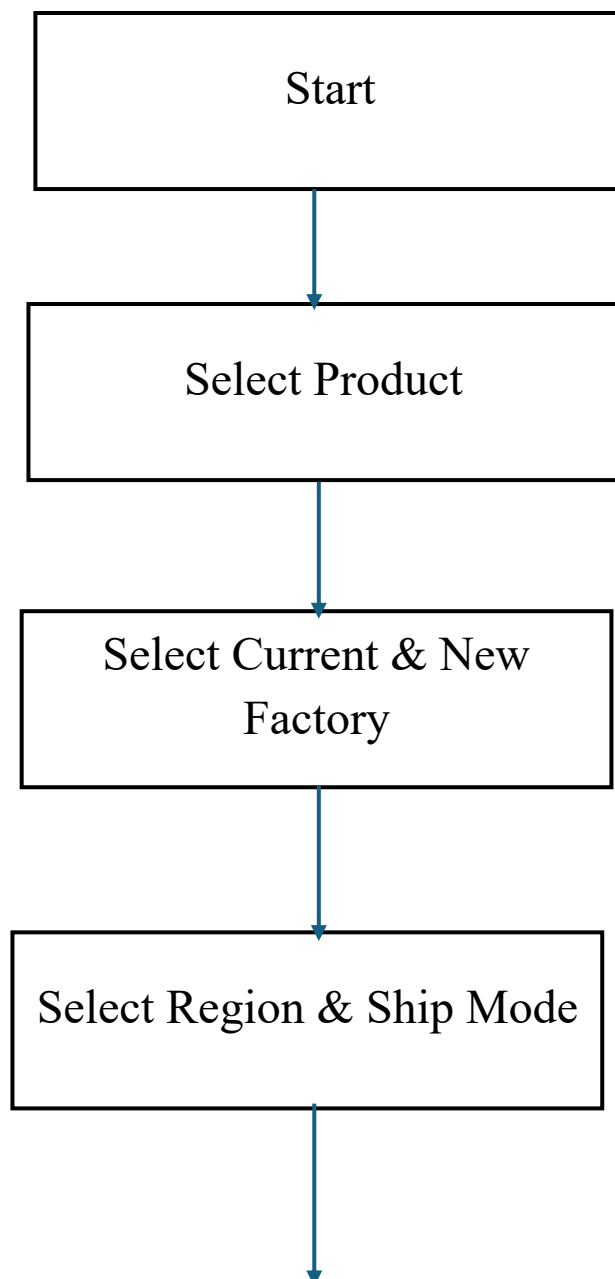


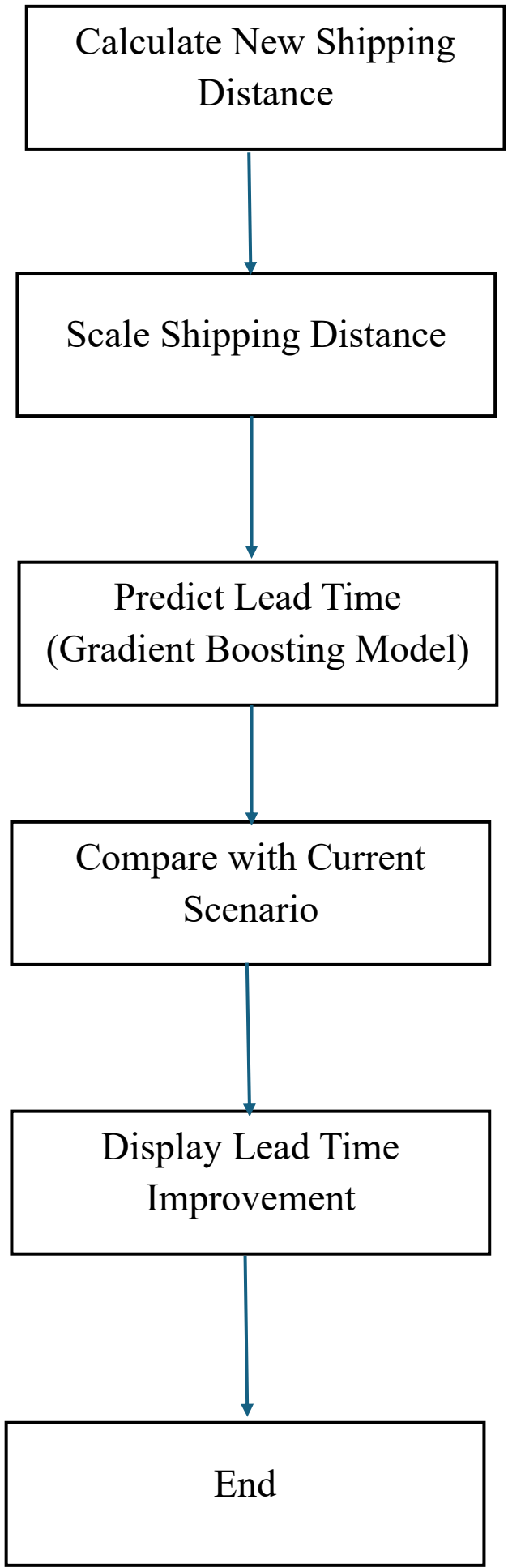


8. What-If Simulation

The What-If Simulation enables users to evaluate the impact of changing the manufacturing factory before implementation. Users select the **Product**, **Current Factory**, **Suggested Factory**, **Region**, and **Ship Mode**, after which the system recalculates the shipping distance for the new factory, scales the distance, and predicts the expected lead time using the trained Gradient Boosting model.

By comparing the predicted lead time with the existing factory assignment, the system provides a realistic estimate of potential improvements, supporting informed and data-driven factory reallocation decisions.





9. Dashboard

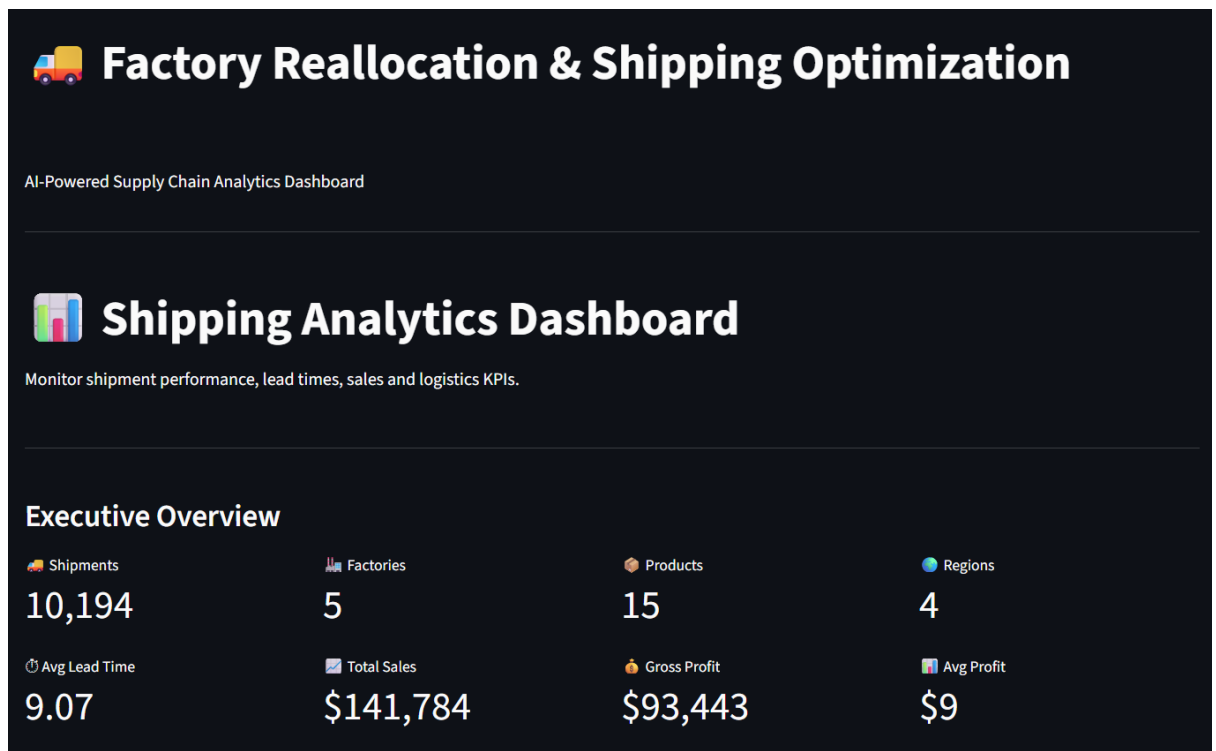
An interactive Streamlit dashboard was developed to visualize supply chain performance and support data-driven decision-making. The dashboard presents key logistics metrics, factory performance, product analysis, regional distribution, optimization recommendations, and a What-If simulation interface. It enables users to monitor operations and evaluate alternative factory allocation strategies in real time.

Dashboard Modules

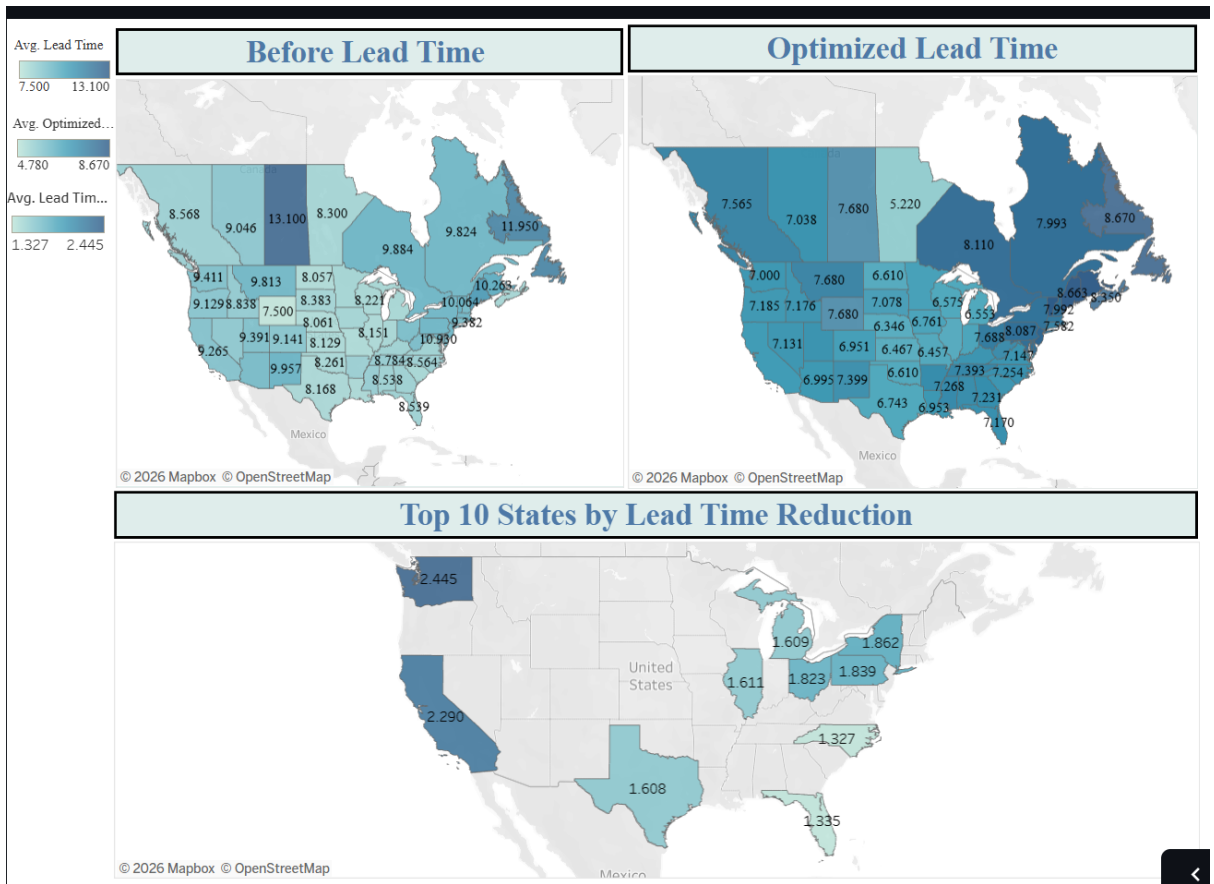
- KPI Cards (Shipments, Sales, Profit, Lead Time)
- Geographic Maps
- Factory Analysis
- Product Analysis
- Recommendation Dashboard
- What-If Simulation

Figures

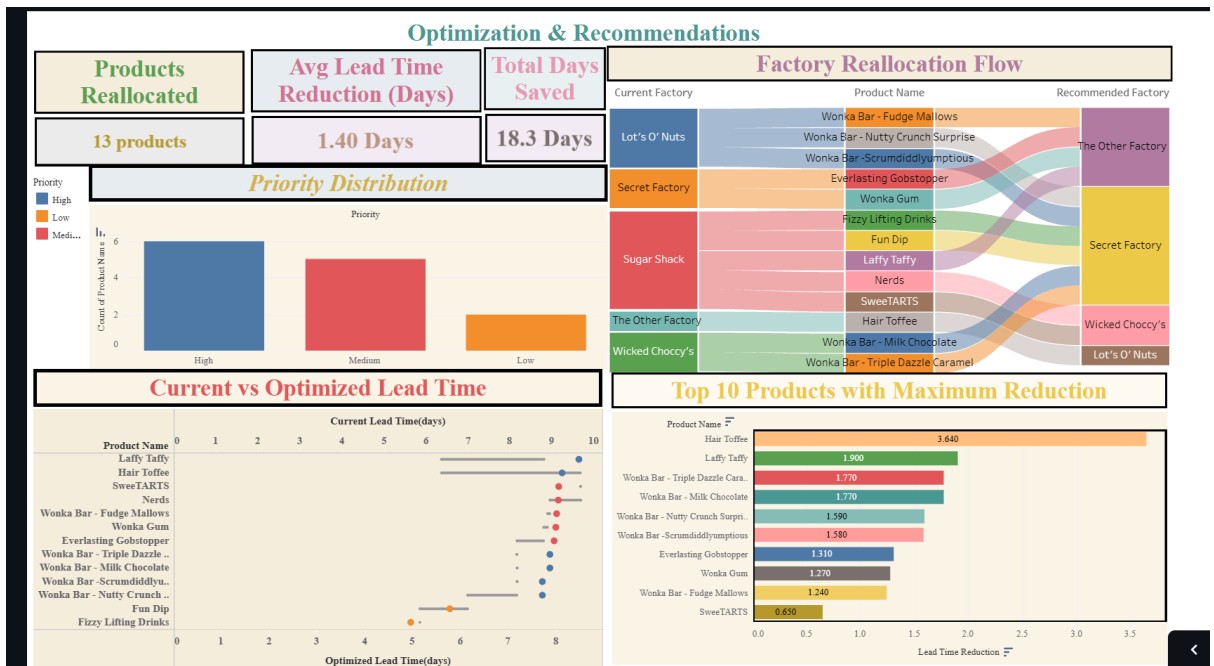
- **Figure 1:** Dashboard Home (KPI Cards)



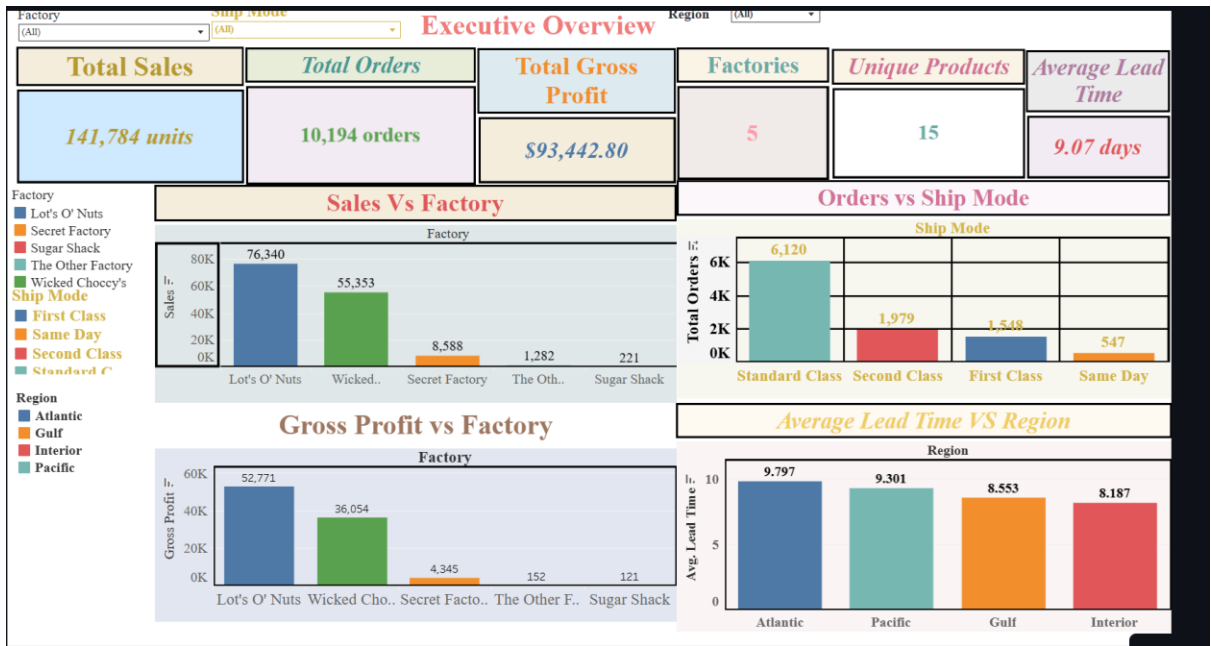
- Figure 2: Geographic Distribution Map



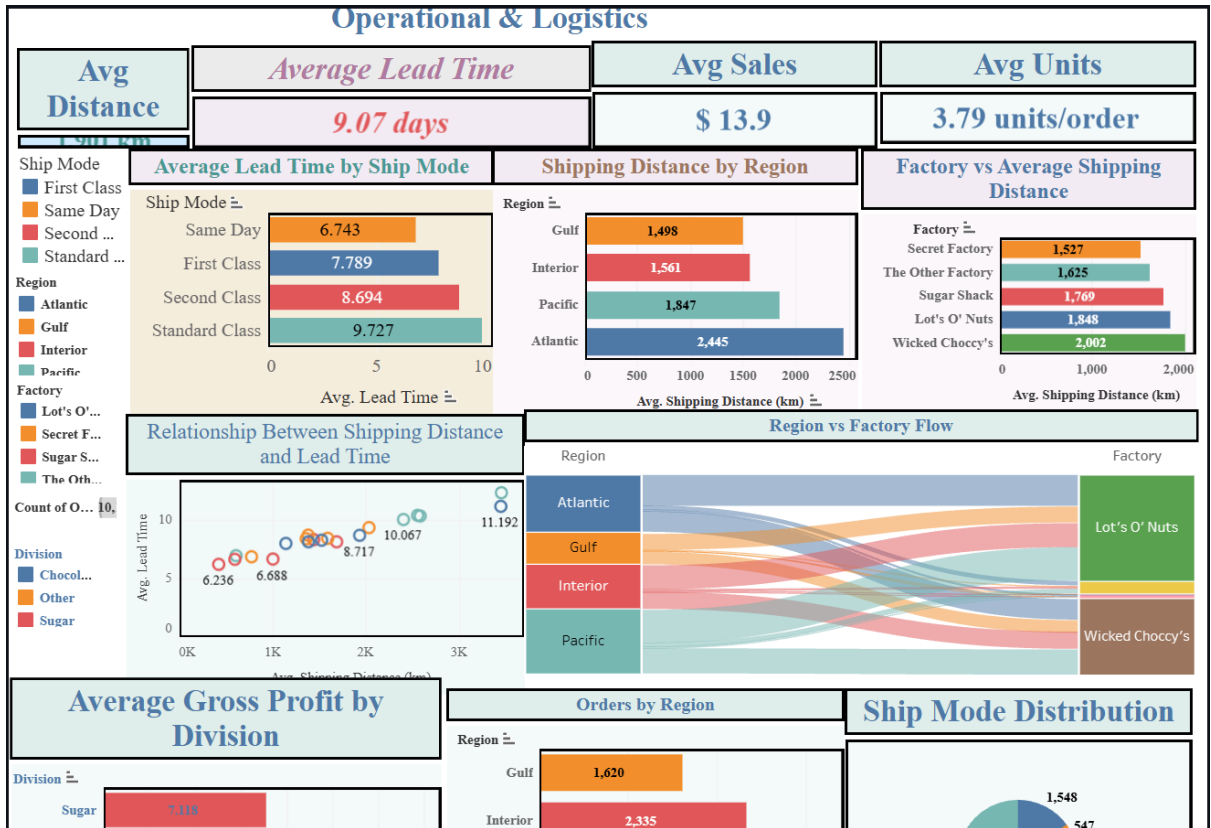
- Figure 3: Factory Performance Analysis



- Figure 4: Product Analysis Dashboard



- Figure 5: Factory Recommendation System



- Figure 6: What-If Simulation Interface



What-If Factory Simulation 00

Simulate factory reallocation and predict the expected shipping lead time before implementation.

Simulation Inputs

Product

Everlasting Gobstopper

Current Factory

Lot's O' Nuts

Suggested Factory

Lot's O' Nuts

Ship Mode

First Class

 Run Simulation

10. Business Impact

The proposed factory reallocation framework improves supply chain efficiency by recommending the most suitable manufacturing facility for each product. Reducing delivery lead time enables faster order fulfillment and enhances customer satisfaction. The optimization process also minimizes shipping distance, which can lower transportation costs and improve overall logistics performance.

The system promotes better factory utilization by distributing production based on predictive insights rather than fixed allocation rules. Additionally, the interactive dashboard and What-If Simulation provide decision-makers with valuable support for evaluating alternative production strategies before implementation.

11. Future Scope

The proposed system can be further enhanced by integrating advanced machine learning models such as **XGBoost** and deep learning techniques to improve prediction accuracy. Real-time data from **ERP systems**, **GPS tracking**, **traffic APIs**, and **weather services** can be incorporated to make predictions more dynamic and responsive.

Future versions may also include **carbon emission optimization**, allowing organizations to minimize environmental impact while maintaining efficient logistics operations.

12.Conclusion

This research demonstrates that **Gradient Boosting** can accurately predict delivery lead time using historical logistics and shipment data. The factory recommendation framework identifies alternative manufacturing locations that can reduce lead time and improve supply chain performance.

The developed Streamlit dashboard enables interactive visualization of logistics data, while the What-If Simulation allows planners to evaluate different factory allocation scenarios before implementation. Together, these components provide a practical decision-support system for data-driven supply chain optimization.

13. References

1. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
2. McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference.
3. Streamlit Inc. *Streamlit Documentation*. <https://docs.streamlit.io>
4. Plotly Technologies Inc. *Plotly Python Documentation*. <https://plotly.com/python/>
5. GeoPy Contributors. *GeoPy Documentation*. <https://geopy.readthedocs.io>
6. Friedman, J. H. (2001). *Greedy Function Approximation: A Gradient Boosting Machine*. Annals of Statistics, 29(5), 1189–1232.
7. Chopra, S., & Meindl, P. *Supply Chain Management: Strategy, Planning, and Operation*. Pearson Education.
8. Christopher, M. *Logistics & Supply Chain Management*. Pearson Education.